

GAIRA V7 — Phase 06

The Validated Chemistry Evidence Layer

Figure report — all 22 figures with captions · Raman only

Status	COMPLETE — 18 of 18 gates pass
Implements	A-19 49-d CSM activation → 16-d Chemistry Evidence
Does not implement	BSV2 (Phase 07) · hierarchical retrieval (Phase 08)
Selected model	D:A_max_idf:lam0.5 (nested CV, modal of 5 folds)
Calibration	temperature (log loss, subject to non-degeneracy floors)
fine-class top-1	0.835 95% CI [0.775, 0.889]
fine-class top-3	0.976 95% CI [0.947, 0.997]
macro-F1	0.793 (classes with ≥5 molecules: 0.826)
balanced accuracy	0.830
ECE / classwise ECE	0.125 / 0.026
replicate consistency	0.947
held-out chemistry novelty	mean AUROC 0.863 over 6 classes — one fails at chance
broken provenance chains	0 of 1125
Frozen inputs	LSM 208482d6... · CSM 0b4aa550... · engine 20d8bd99... — all verified
Completed	2026-08-06 23:31:15 UTC

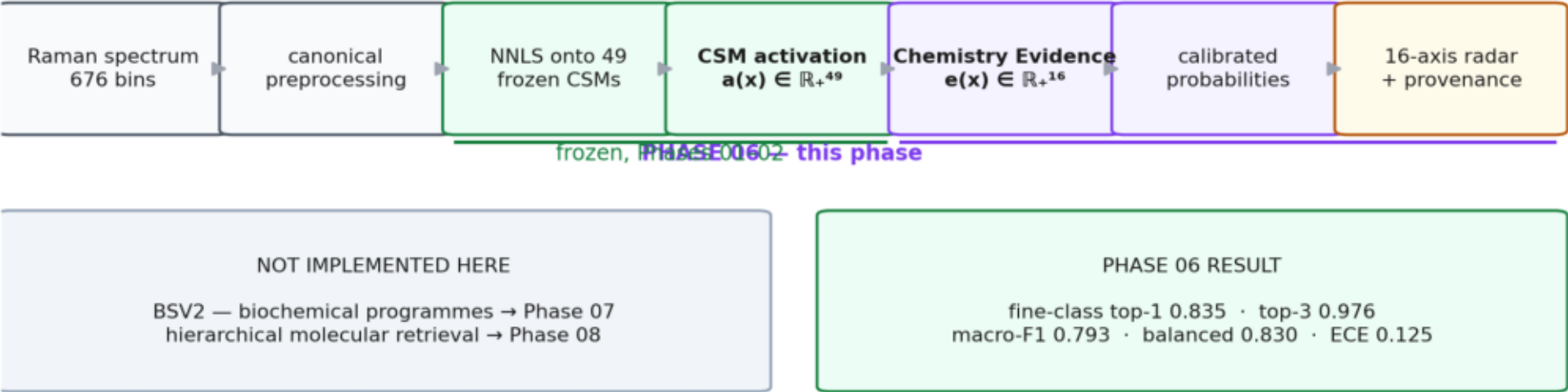
Committed sources of record: reports/PHASE_06_CHEMISTRY_EVIDENCE_LAYER.md and reports/PHASE_06_SCIENTIFIC_AUDIT.md
The evidence vector is support within the chemistry represented by the Raman reference atlas. It is NOT a concentration, NOT a composition, and NOT a mixture statement. Every spectrum in this phase is a pure Raman reference.

Figures

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Figure 01

Figure 1 · The GAIRA V7 pipeline through Phase 06



The GAIRA V7 pipeline through Phase 06. Everything left of the purple rule is frozen; Phase 06 adds the 16-dimensional Chemistry Evidence layer and its calibration. BSV2 and hierarchical retrieval are explicitly out of scope.

Figure 2 · The exact chemistry-class inference mathematics

What Phase 05 actually computed (traced to source, not what Phase 06 computes)

engine.py:86-89 · run_phase05.py::topk_class

$$e_c(x) = \max_{i \in c} \cos(a(x), r_i)$$

one nearest molecule decides the class
no class-size correction · no calibration
no probability · not a vector

selected by nested CV: D:A_max_idf:lam0.5

$$e_c(x) = w_c \cdot \max_{i \in c} \cos(a(x), r_i) \cdot b_{\beta(c)}(x)^\lambda$$

w_c inverse-frequency size correction
 $b_{\beta(c)}$ SOFT broad-superclass routing, $\lambda=0.5$
evidence for all 16 classes, then calibrated

reproduced bit-for-bit: top-1 0.845310, top-3 0.970667, macro-F1 0.806783, balanced 0.796612

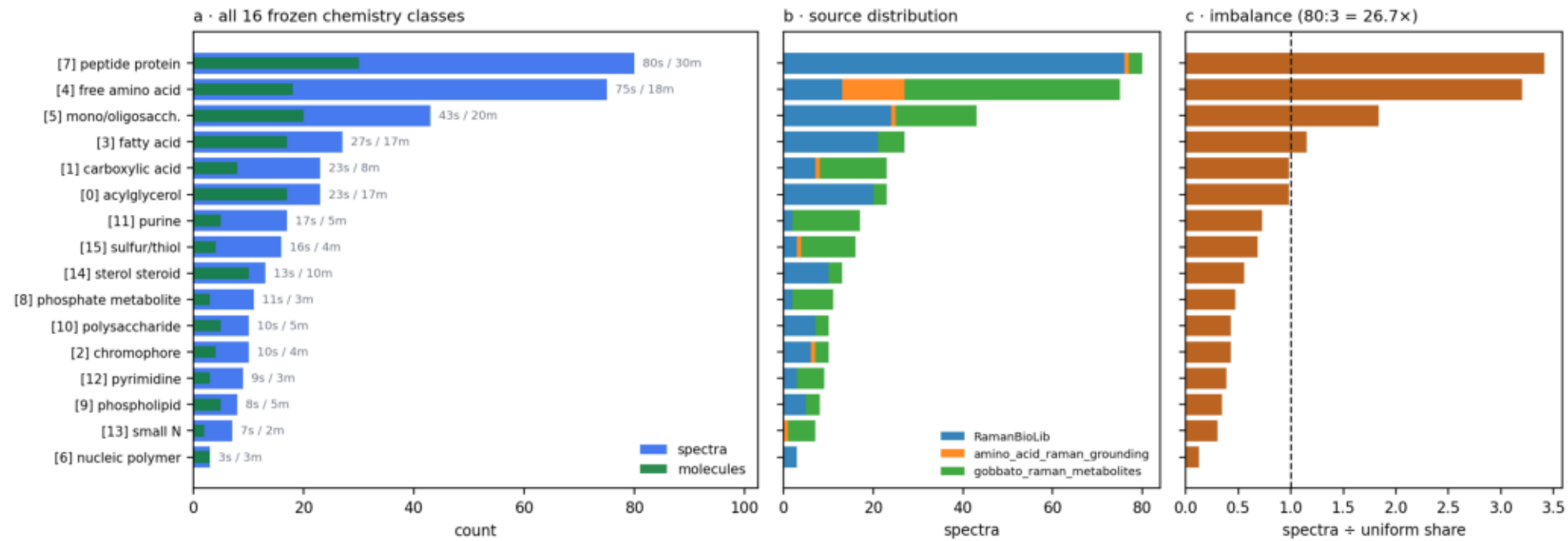
Why the difference matters

- Phase 05 returned a LABEL. Phase 06 returns a 16-dimensional continuous vector: every class carries evidence, ambiguity is visible, and the second-best chemistry is a number rather than an absence.
- The size correction stops a 80-spectrum class outscoring a 7-spectrum class by having more chances to contain a near neighbour.
- Broad routing is SOFT — multiplicative and strictly positive — so a fine class stays reachable when its superclass is not top-1. A hard filter would make a broad error unrecoverable.
- The evidence is exactly decomposable: e_c is an explicit function of named molecule similarities, each an inner product of named CSM activations.

The exact inference mathematics, traced to source rather than to report prose. Phase 05 returned a label from one nearest molecule; Phase 06 returns a 16-dimensional vector with a class-size correction and soft broad-superclass routing, then calibrates it.

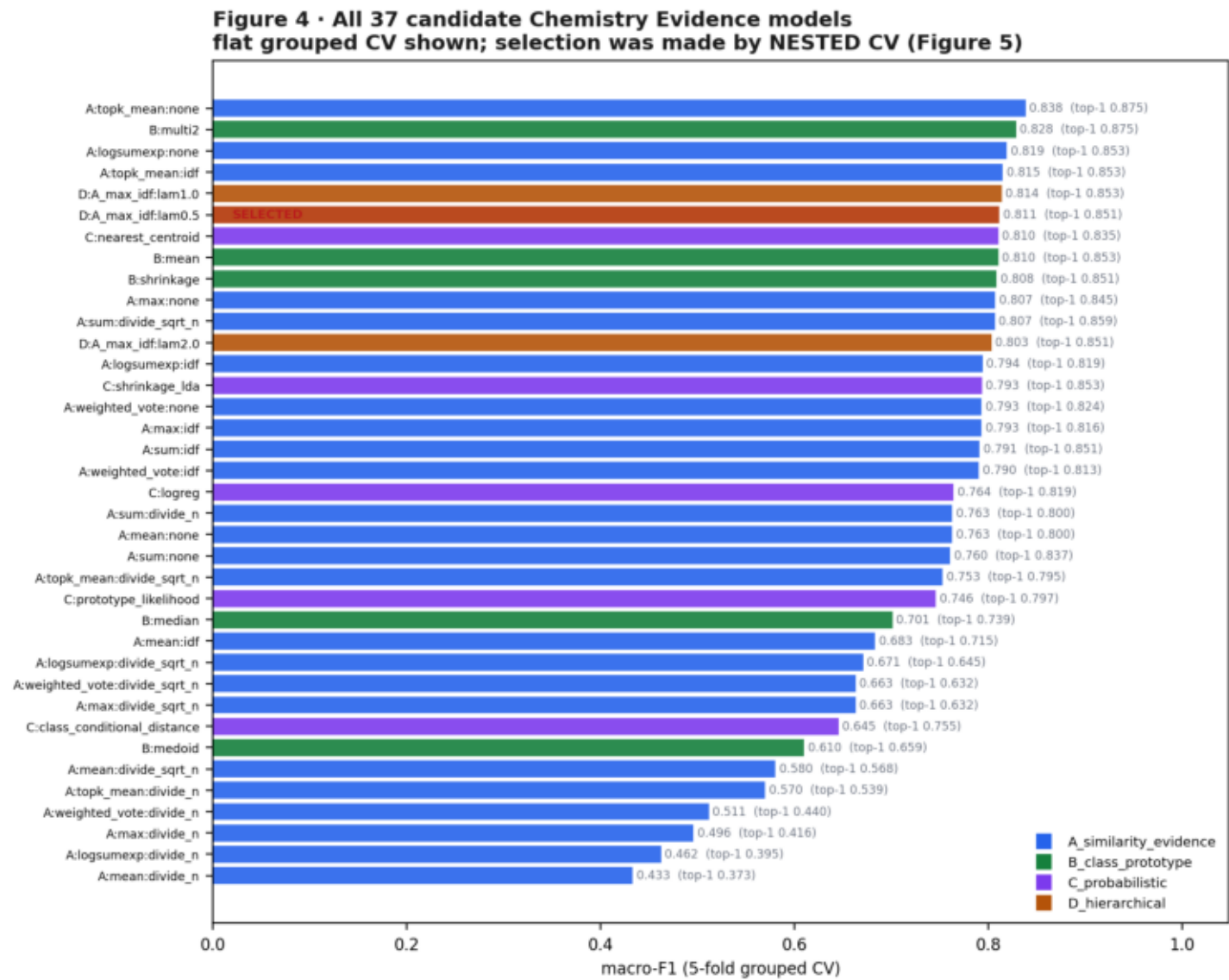
Figure 03

Figure 3 · The frozen 16-class corpus — 375 spectra, 154 molecules



All 16 frozen chemistry classes printed explicitly, with molecules, spectra, source distribution and imbalance. The largest class holds 80 spectra and the smallest 3 — a 26.7x ratio that §8 of the report shows is the dominant driver of per-class F1.

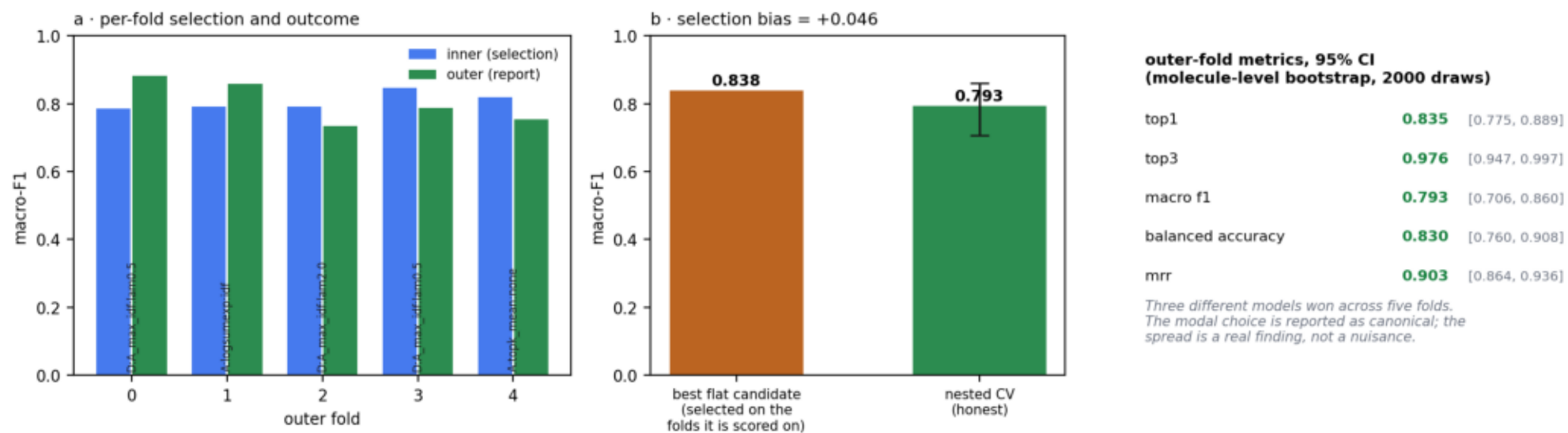
Figure 04



All 37 candidate models across four families under flat grouped CV. This ranking is shown for the record only: selection was made by nested CV, and Figure 5 shows the 0.045 macro-F1 difference between the two.

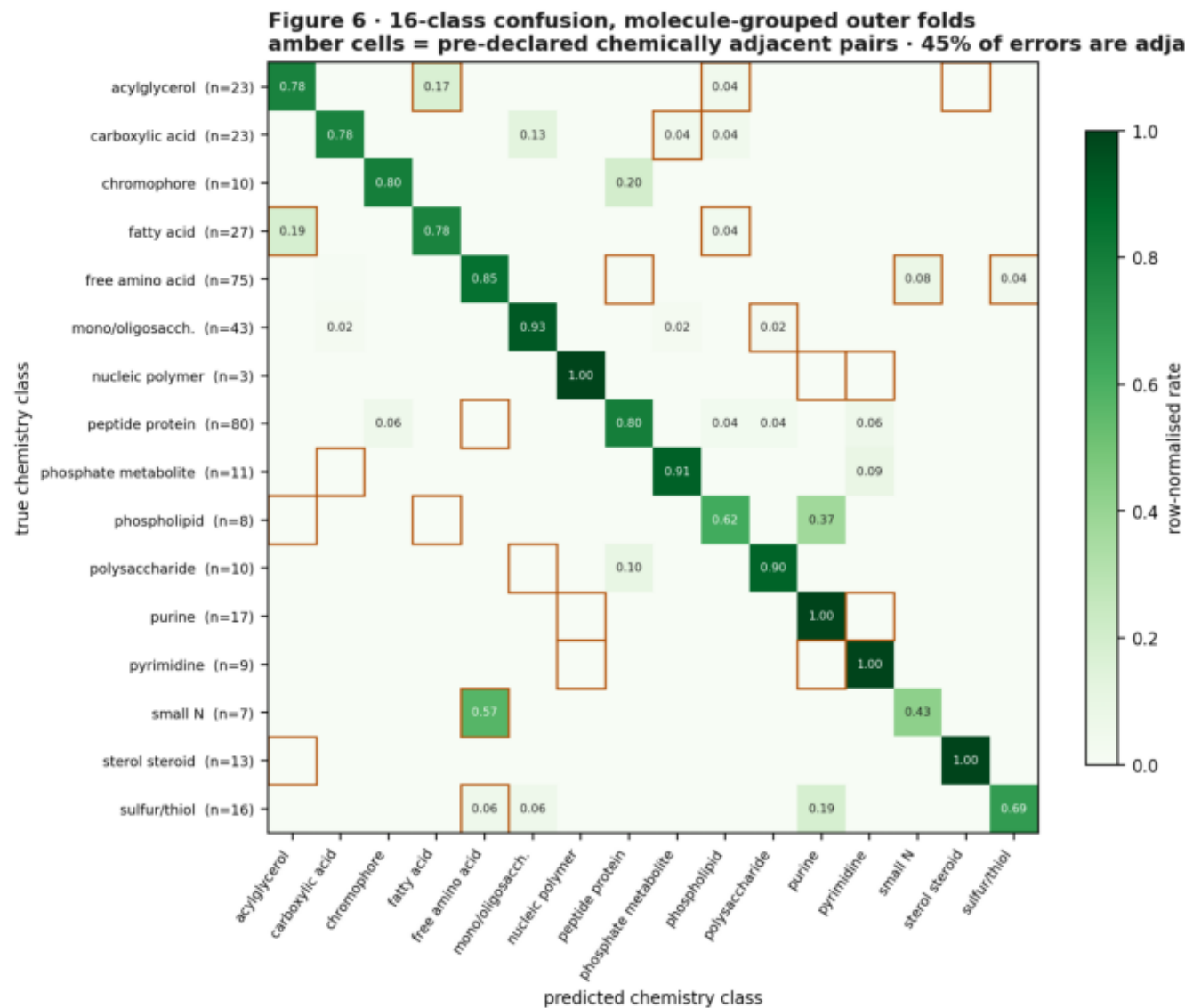
Figure 05

Figure 5 · Nested molecule-grouped cross-validation



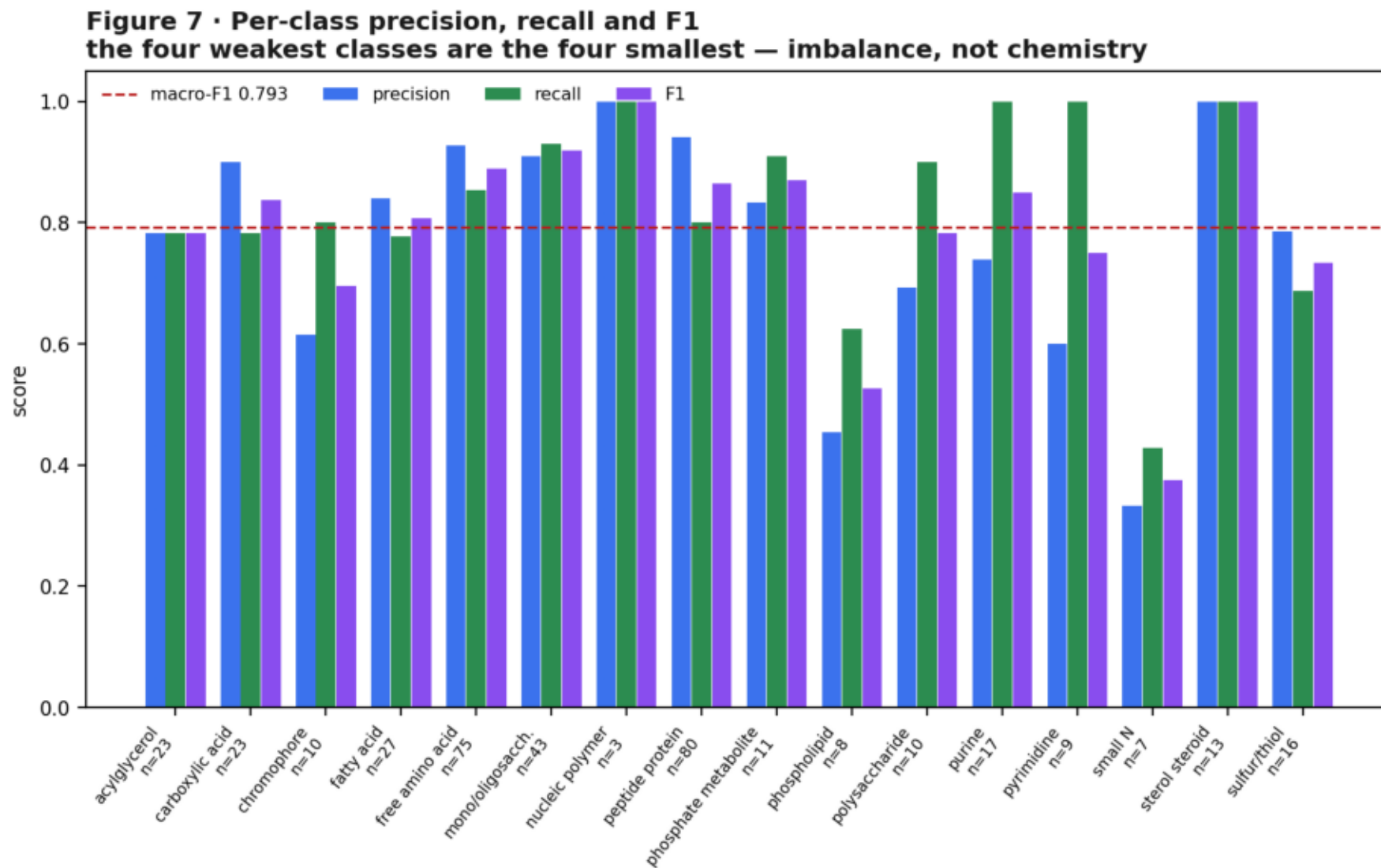
Nested molecule-grouped cross-validation. Three different models won across five folds; the flat-vs-nested gap of 0.045 is exactly the selection bias that nested CV exists to remove.

Figure 06



16-class confusion on molecule-grouped outer folds. Amber cells mark pre-declared chemically adjacent pairs — 45% of errors land on them against a 11% chance rate.

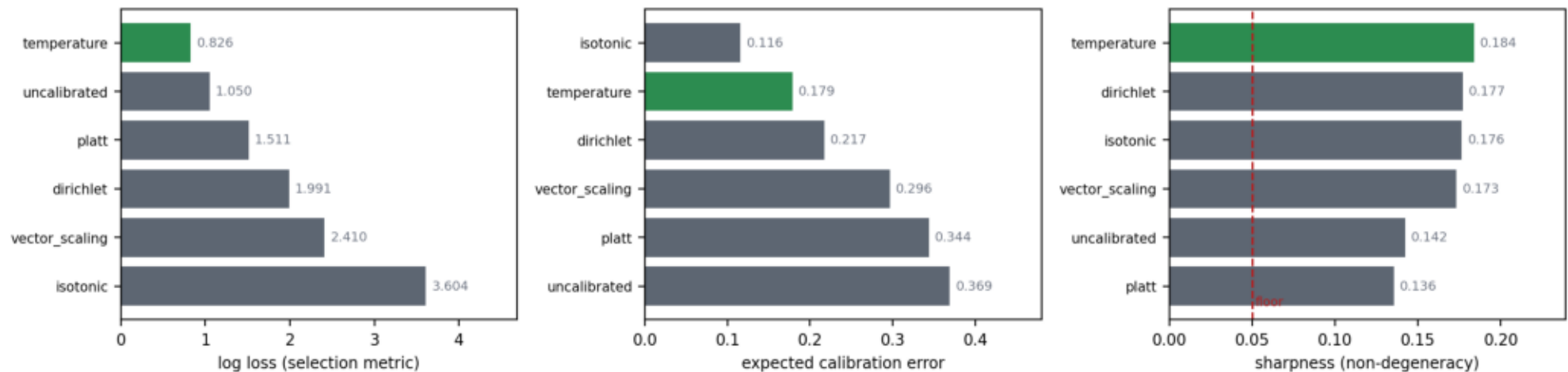
Figure 07



Per-class precision, recall and F1. The four weakest classes are the four smallest, which is a corpus property rather than a modelling defect.

Figure 08

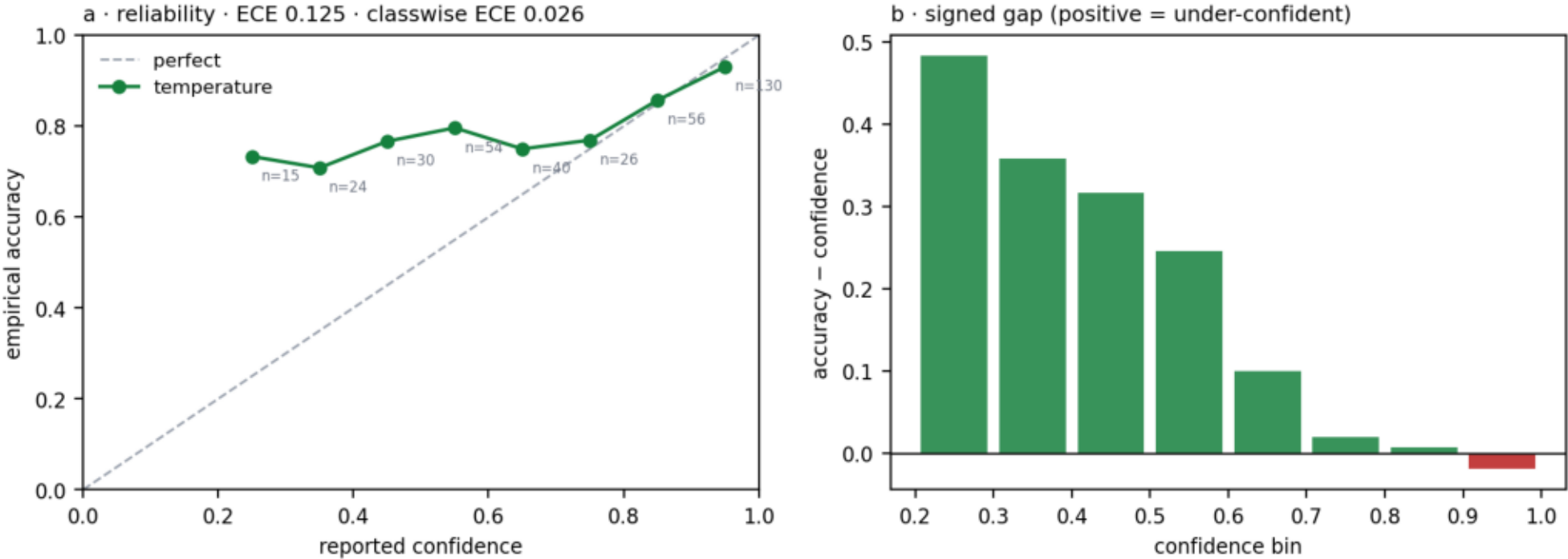
Figure 8 · Calibration benchmark — selection on log loss, subject to pre-registered non-degeneracy floors
isotonic wins ECE (0.116) and loses log loss (3.604): per-class calibration then renormalised destroys the joint likelihood



Calibration benchmark. Isotonic regression wins ECE and loses log loss by a factor of four — per-class calibration then renormalised fixes the marginals and destroys the joint likelihood. Selecting on ECE would have chosen it.

Figure 09

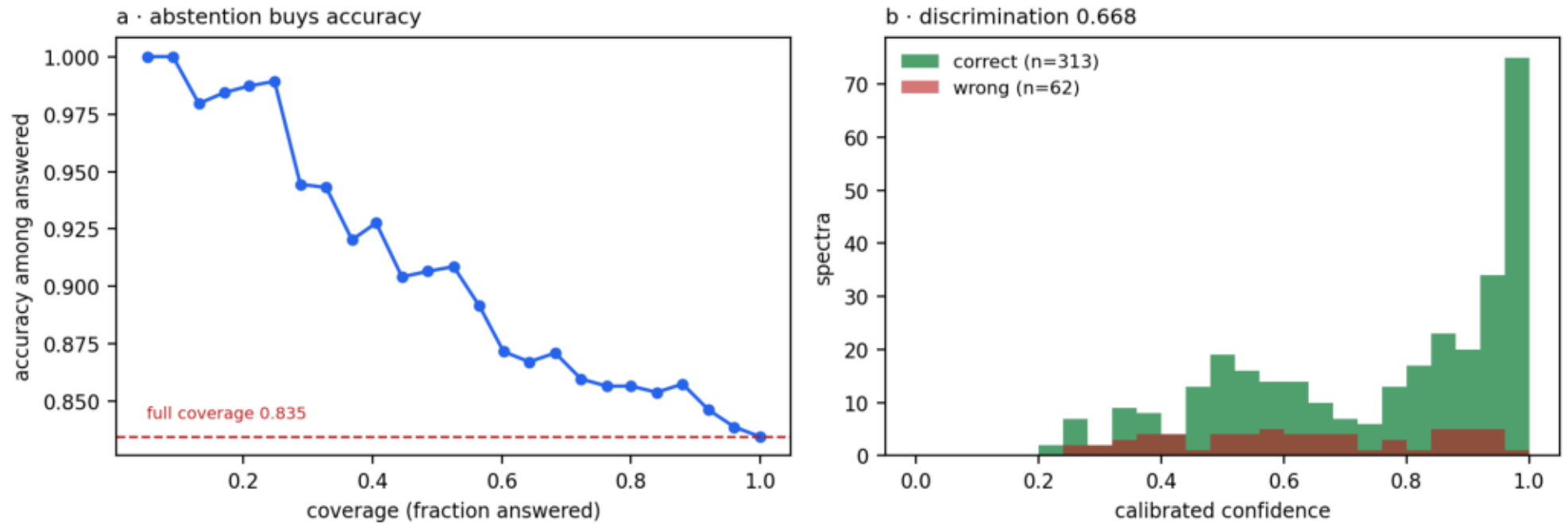
Figure 9 · Reliability of the calibrated Chemistry Evidence



Reliability of the selected temperature calibrator. Classwise ECE of 0.026 is the reassuring number: small classes are not systematically miscalibrated, which top-label ECE alone would hide.

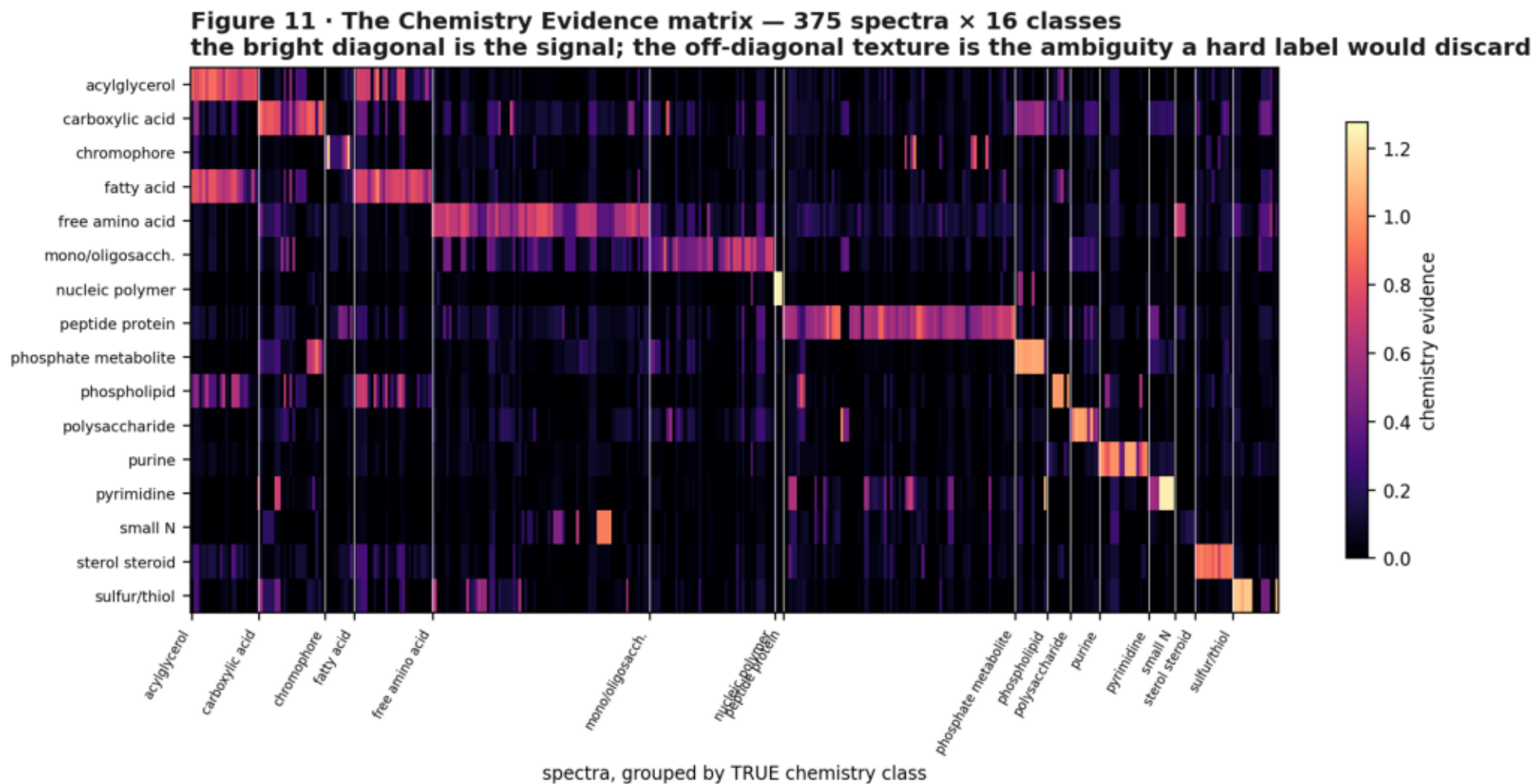
Figure 10

Figure 10 · Selective accuracy versus coverage



Selective accuracy versus coverage, and the confidence distribution split by correctness. Discrimination of 0.668 is modest and Figure 16 shows the consequence.

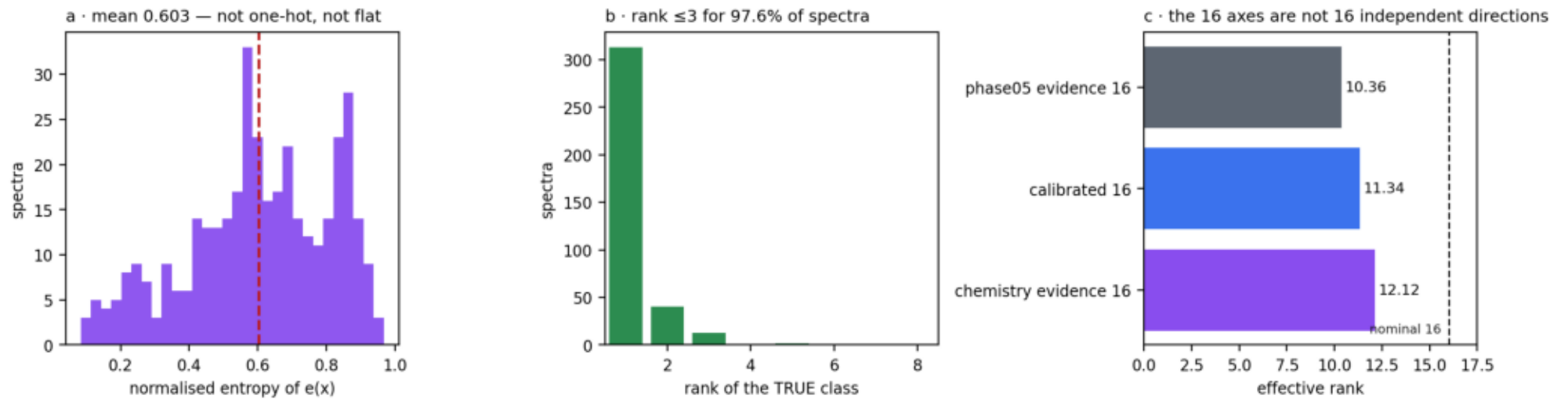
Figure 11



The full 375 x 16 evidence matrix, spectra grouped by true class. The bright diagonal is the signal; the off-diagonal texture is the ambiguity a hard label discards.

Figure 12

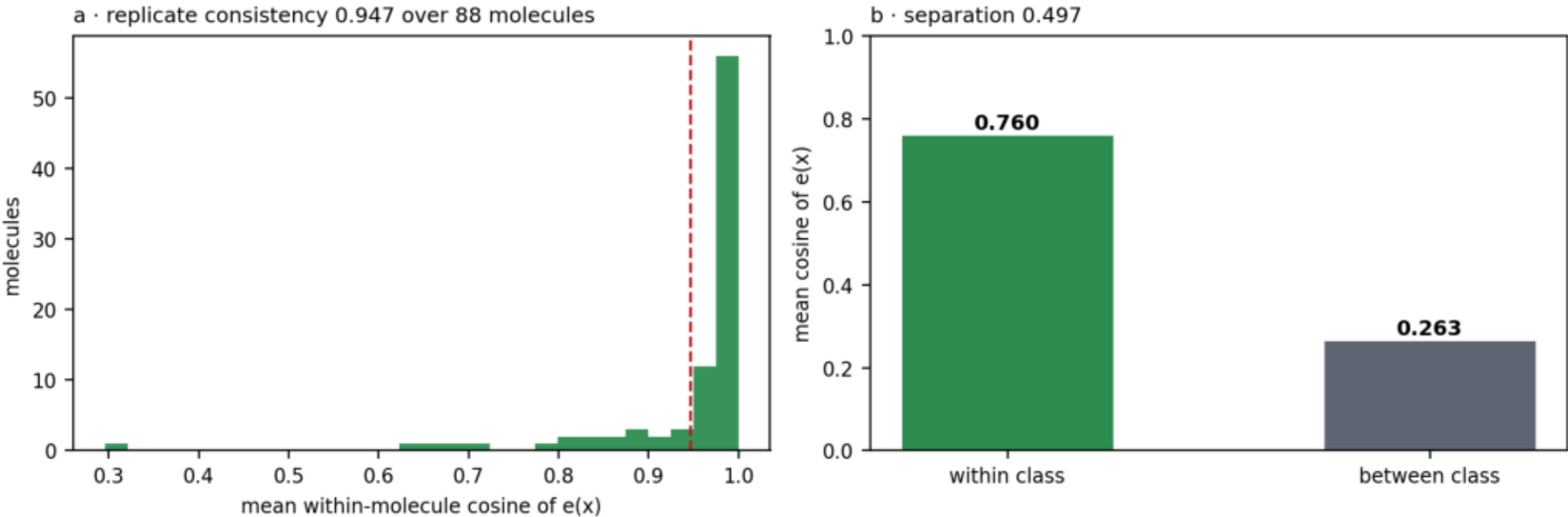
Figure 12 · Effective rank, entropy and true-class rank



Entropy 0.603, true-class rank, and effective rank 12.12 of a nominal 16 — the three numbers that together establish this is a genuinely soft, multi-dimensional representation rather than a disguised classifier.

Figure 13

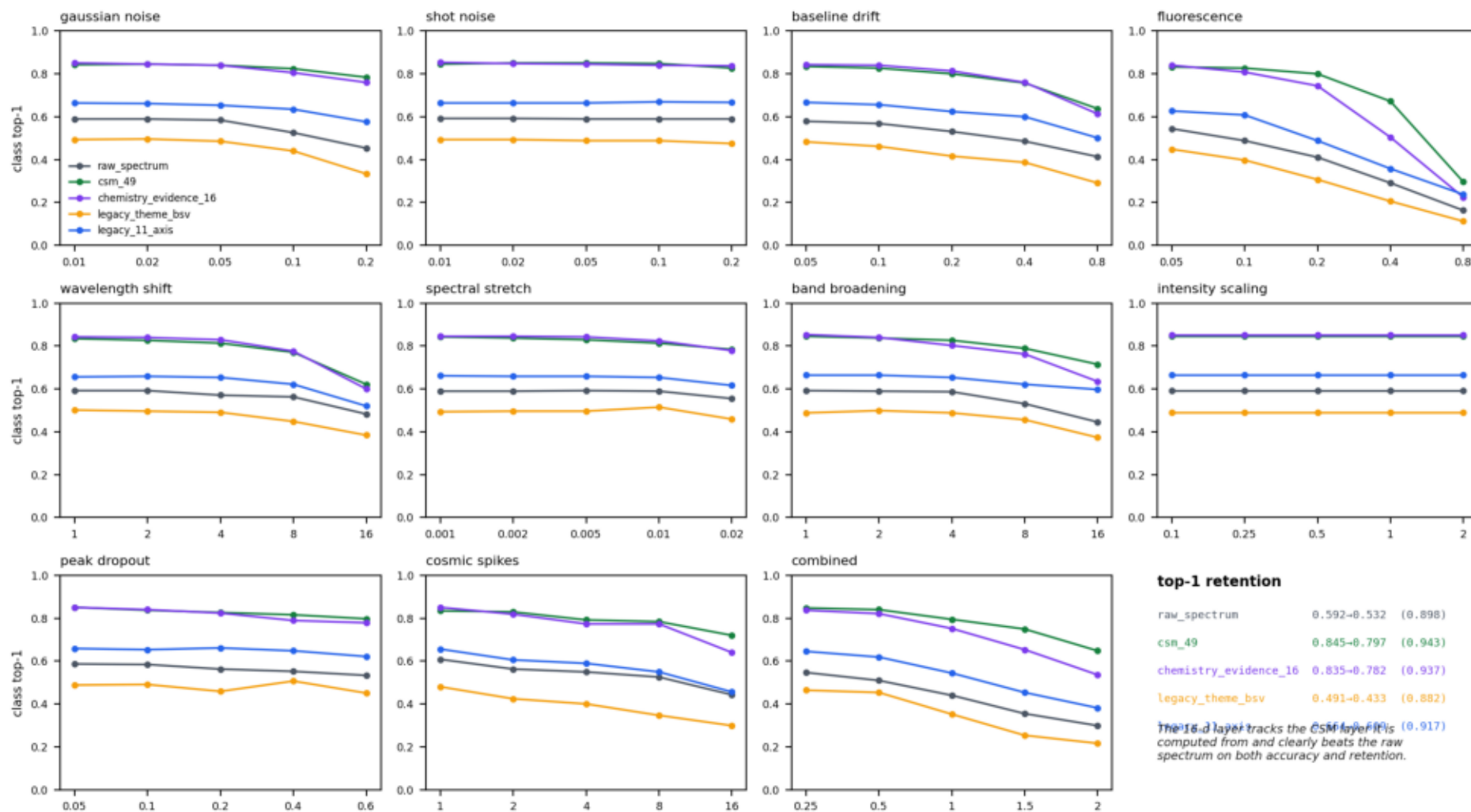
Figure 13 · Replicate and within-class consistency of the evidence vector



Replicate consistency 0.947 and a within/between-class separation of 0.497 on the evidence vector itself.

Figure 14

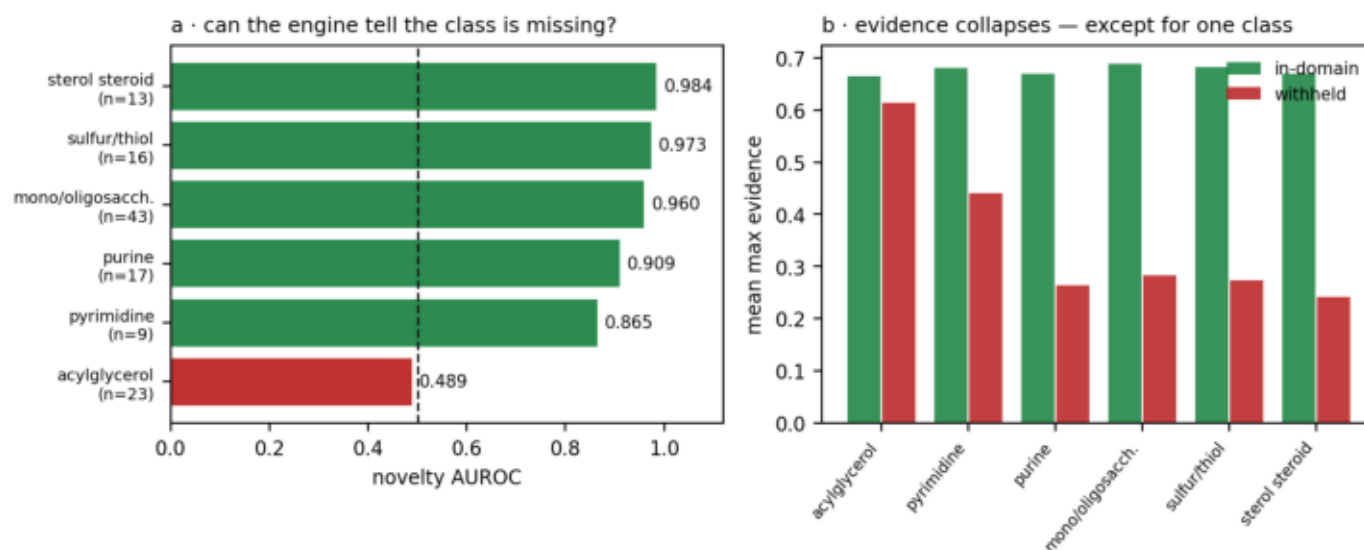
Figure 14 · Raman perturbation robustness — 11 perturbations × 5 levels



Eleven Raman perturbations at five levels each. The 16-d layer tracks the CSM layer it is computed from (retention 0.937 vs 0.943) and clearly beats the raw spectrum on both accuracy and retention.

Figure 15

Figure 15 · Held-out chemistry novelty — an entire class withheld from the atlas (Raman only)



THE FAILURE, STATED PLAINLY

acylglycerol scores AUROC 0.489 — no better than chance — and abstains on 0% of its spectra. Its mean max evidence barely falls (0.614 vs 0.665 in-domain).

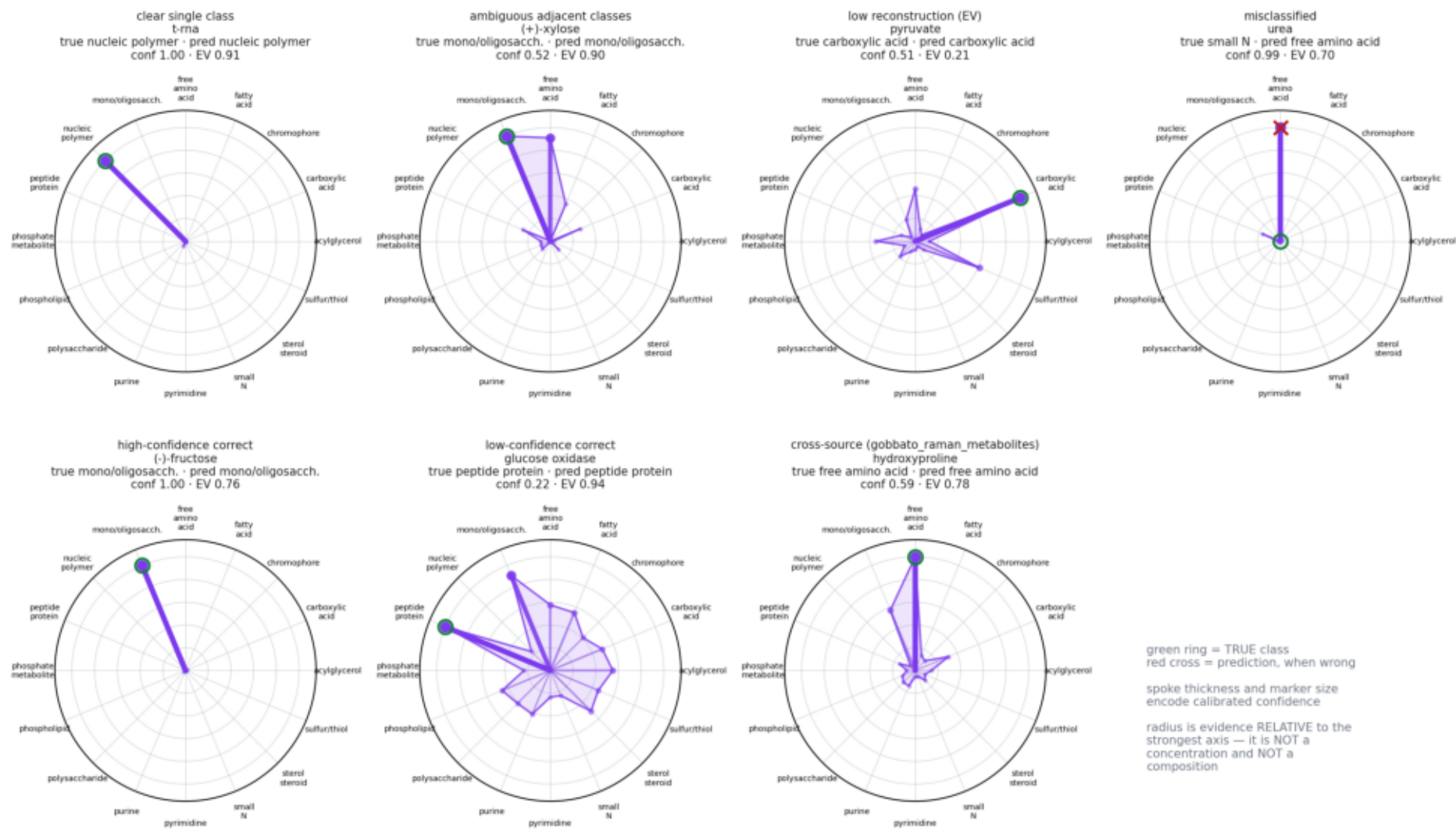
The reason is chemical, not computational: with acylglycerols withheld, fatty acids remain, and a triacylglycerol's Raman spectrum is dominated by the same acyl-chain motifs. The engine is not wrong that the chemistry is present — it is wrong that the chemistry is REPRESENTED.

Novelty detection works when the withheld chemistry has no close represented neighbour, and fails when it does. That is the honest boundary of this capability.

Held-out chemistry novelty — an entire class withheld from the atlas. Five of six classes are detected; acylglycerol fails at chance because fatty acids remain and share the acyl-chain motifs that dominate its spectrum.

Figure 16

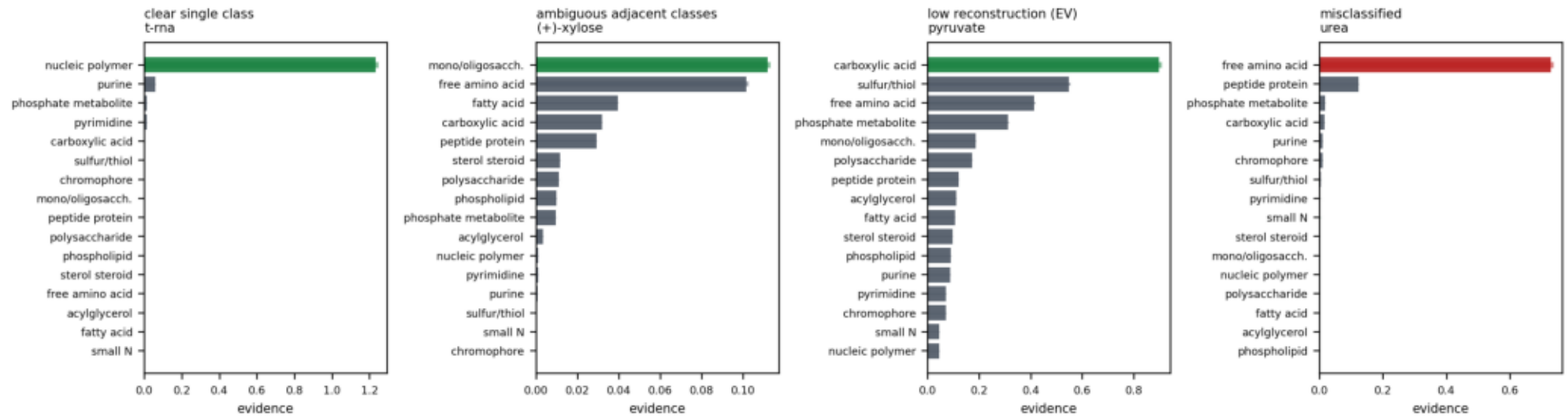
Figure 16 • 16-axis Chemistry Evidence radars — seven representative cases



Sixteen-axis radars for seven cases chosen by rule rather than by eye. Note urea, misclassified as a free amino acid at confidence 0.99 — calibration reduces average over-confidence, it does not catch a confidently wrong answer.

Figure 17

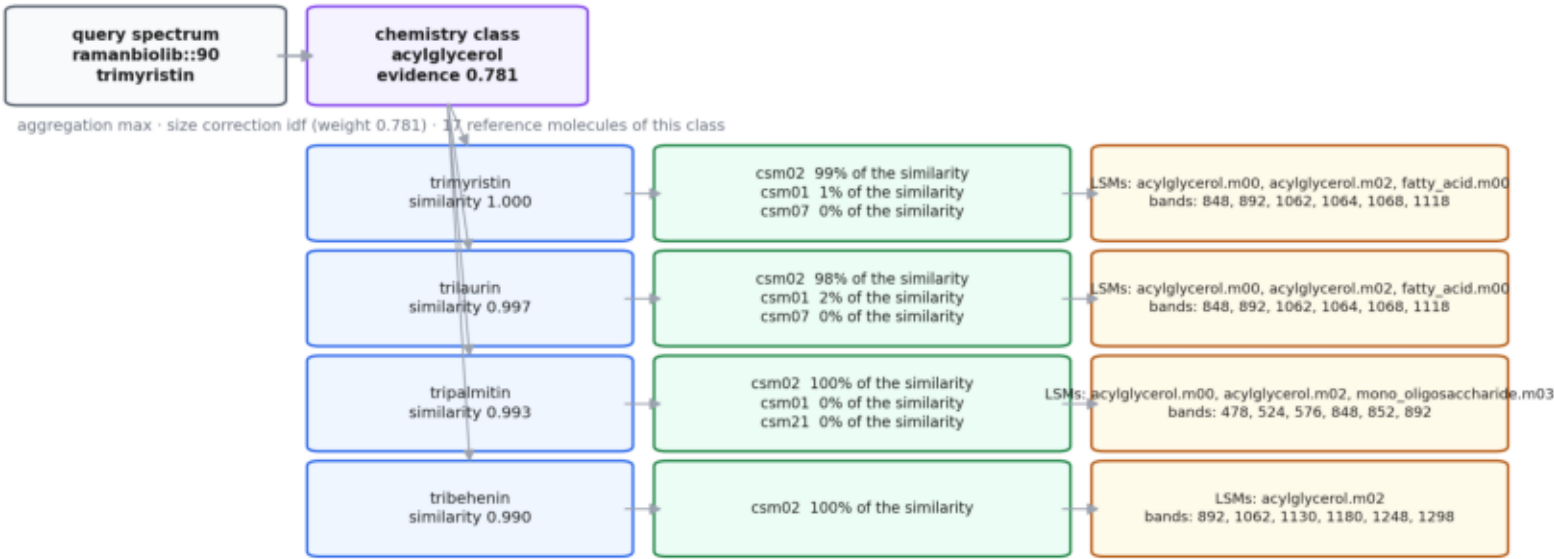
**Figure 17 • Ordered-bar alternative to the radar — same numbers, no angular distortion
green = true class • red = prediction when wrong • bar-line weight encodes confidence**



The ordered-bar alternative. Sixteen spokes are cluttered and a filled polygon invites a composition reading; the same numbers as bars carry neither problem.

Figure 18

Figure 18 · Provenance waterfall — chemistry class → molecules → CSMs → LSMs → spectra

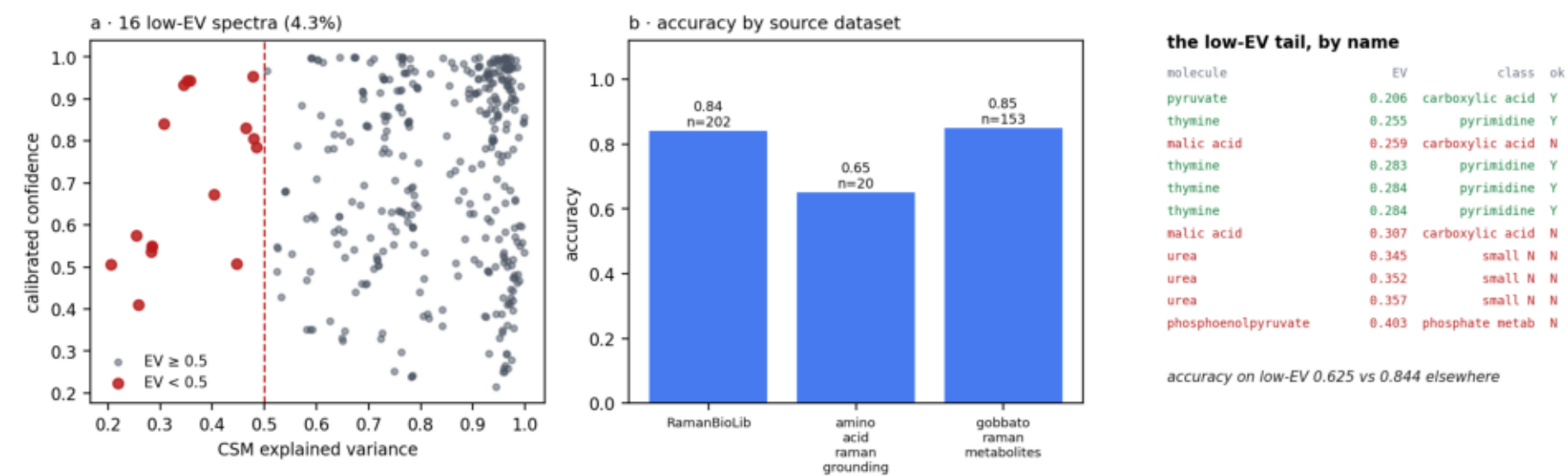


Exact decomposition: True - 1125 chains verified against the frozen registries - 0 broken. Every similarity is an inner product of the query's CSM activation with a named reference activation, so the listed CSMs sum back to the number shown.

Provenance waterfall. The decomposition is mathematically exact for the selected family: 1,125 chains verified against the frozen registries, none broken.

Figure 19

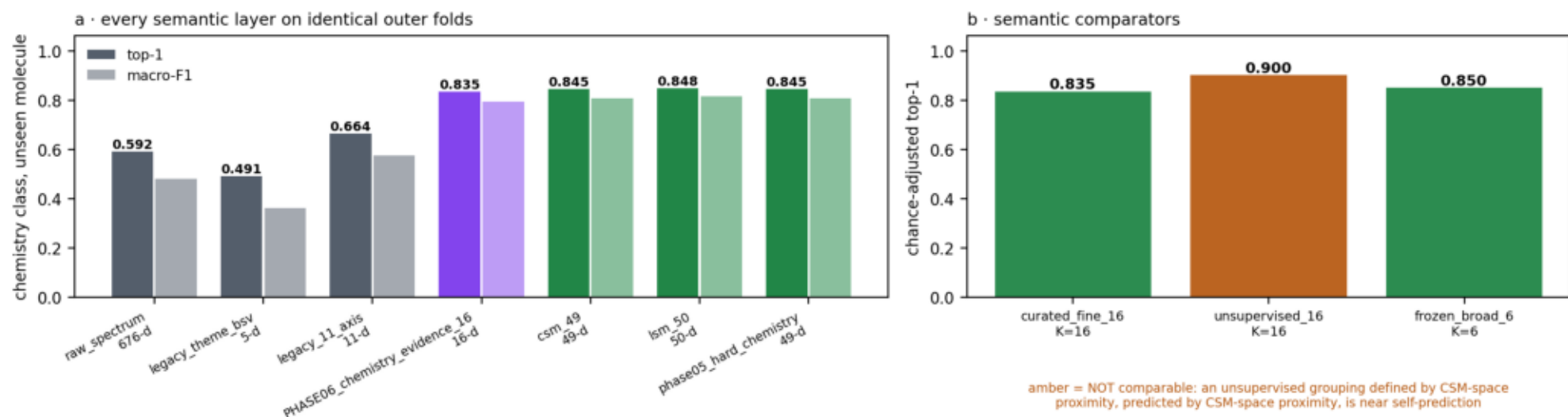
Figure 19 · Low-EV cases and failure analysis



The low-EV tail by name, accuracy by source, and the relationship between reconstruction quality and confidence. Low EV degrades accuracy to 0.625 from 0.844 but does not destroy it.

Figure 20

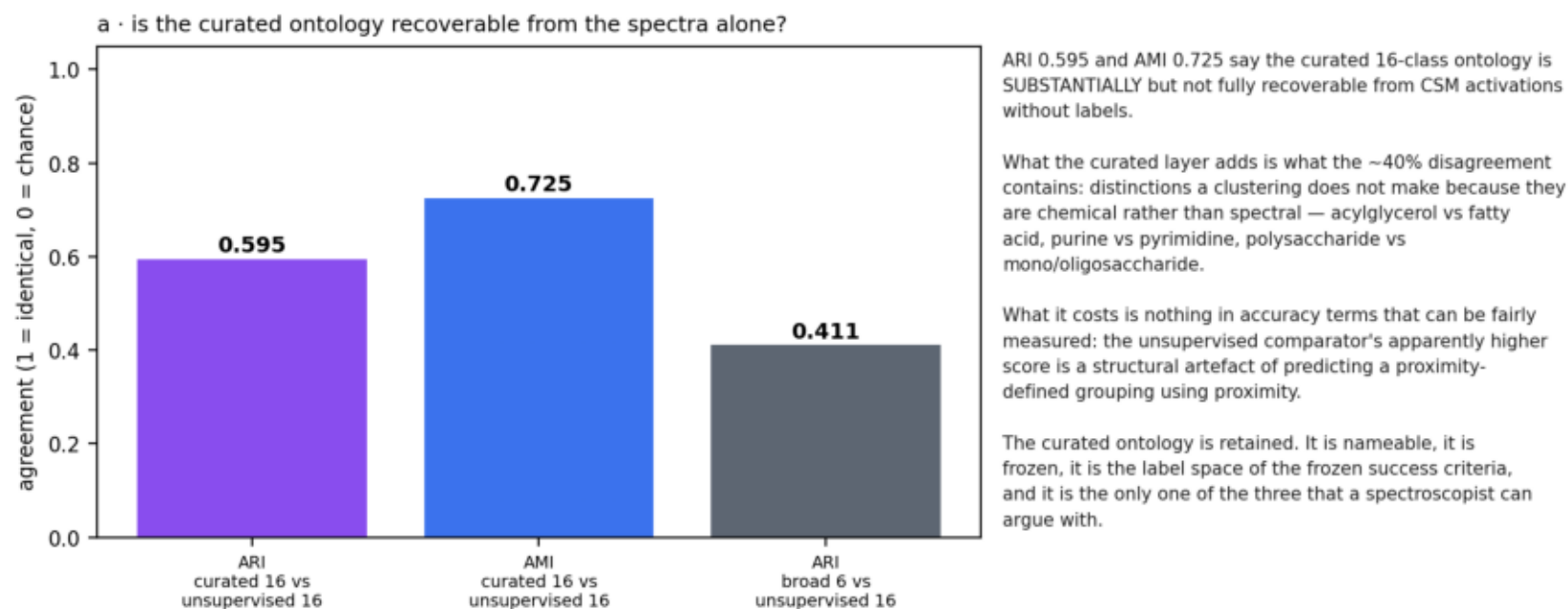
Figure 20 · Comparison with every legacy semantic layer, and with unsupervised grouping



Every semantic layer on identical outer folds, and the semantic comparators. The unsupervised bar is amber because it is not a fair comparison — predicting a proximity-defined grouping using proximity is near self-prediction.

Figure 21

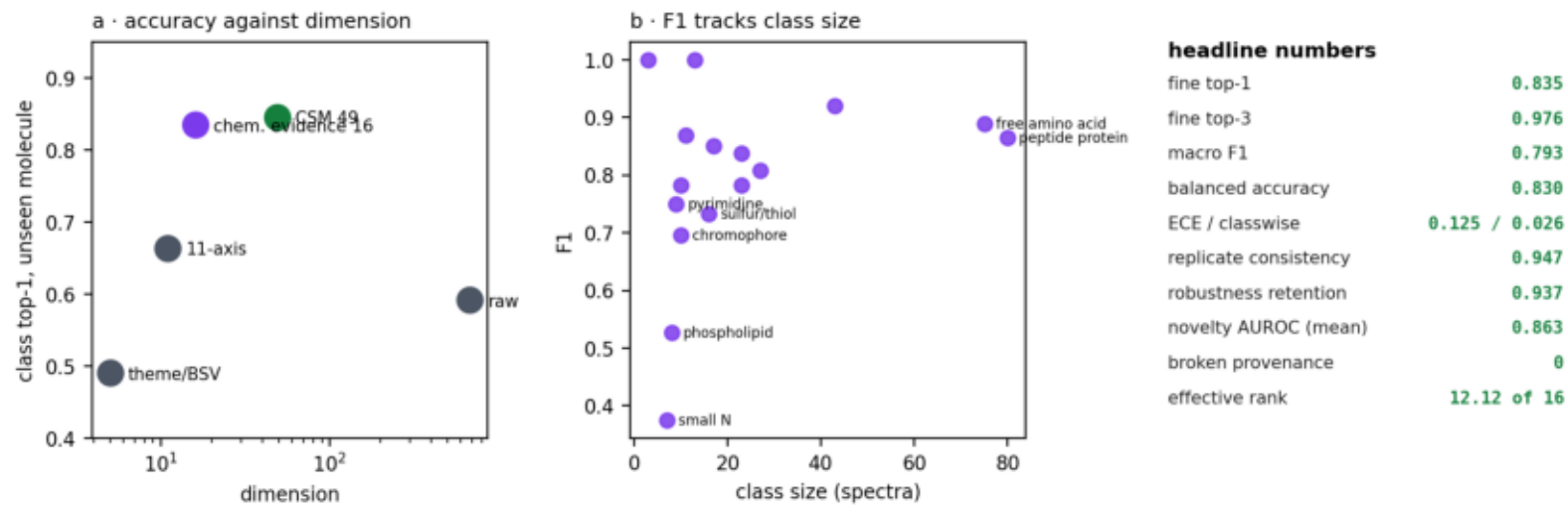
Figure 21 · Curated ontology versus unsupervised semantic comparator



Curated ontology versus unsupervised grouping. ARI 0.595 and AMI 0.725 say the curated partition is substantially but not fully recoverable from spectra alone; the disagreement is where a curated layer earns its place.

Figure 22

Figure 22 · Phase 06 architecture summary



G1 frozen fingerprints verified	G2 no upstream refitting	G3 nested molecule-grouped validation used	G4 fine-class top-1 >= 0.80	G5 fine-class top-3 >= 0.95
G6 macro-F1 >= 0.75	G7 chemistry evidence matches CSM class top-1 within 0.02	G8 exceeds legacy Theme/BSV top-1 by >= 0.20	G9 replicate consistency >= 0.90	G10 robustness retention >= raw Raman
G11 calibration informative and non-degenerate	G12 no broken provenance chains	G13 radar generated from the 16-d chemistry vector only	G14 no manual 11-axis map in canonical Phase 06 inference	G15 no SERS or cross-modality data used
G16 held-out chemistry novelty evaluated honestly	G17 Phase 06 artifacts deterministic on rerun	G18 every pre-registered candidate and calibrator actually ran		

18 of 18 gates pass · selected model D:A_max_idf:lam0.5 · calibration temperature · Raman only, no cross-modality data